

07_phase2_synthesis

HXC-159 Phase 2 — Specification + implementation-plan synthesis

Field	Value
Document	07_phase2_synthesis.md
Work item	HXC-159 (Task, In-progress, High)
Phase	2 — SpecKit specification + phase/task/subtask implementation plan
Revision	1
Last modified	2026-07-29
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Status summary	Specification and plan authored at specs/001-helixskills-incorporation/. 63 tasks across 13 phases in 2 concurrent tracks. 29 register risks → 29 mitigations → task coverage , plus 3 new risks found during planning. 9 UNCONFIRMED: items carried, 2 resolved during this phase. Nothing implemented.
Anchors	§11.4.6 · §11.4.8 · §11.4.35 · §11.4.74 · §11.4.108 · §11.4.118 · §11.4.122 · §11.4.169 · §11.4.201 · §11.4.224 · CONST-051(C) · CONST-059

Phase 2 does not re-derive Phase 1. Its value is synthesis: turning the six Phase-1 documents into a specification with per-feature dispositions, a sequenced plan, and a risk → mitigation → task map. Where Phase 2 *corrected* a Phase-1 or brief-level claim, the correction is recorded in §2 rather than silently absorbed.

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1. Deliverables

Artefact	Path	Contents
Specification	specs/001-helixskills-incorporation/spec.md	5 prioritised user stories · 50 functional + 4 non-functional requirements · all 57 upstream capability rows dispositioned · all 23 constitution-derived extensions dispositioned · all 7 Bridge layers dispositioned · 14 success criteria · 9 UNCONFIRMED: items
Implementation plan	specs/001-helixskills-incorporation/plan.md	Technical context · constitution check · structure decision · 13-phase / 2-track architecture with per-phase rationale · finding-by-finding handling · risk → mitigation → gate → mutation table · test strategy · 22-entry runtime-signature registry · complexity tracking · doc deliverables
Task breakdown	specs/001-helixskills-incorporation/tasks.md	63 tasks , each with scope / §11.4.169 test types / §11.4.108 runtime signature / acceptance evidence / risks, decomposed into ~230 subtasks · dependency graph · parallel-execution guidance
SpecKit constitution pointer	.specify/memory/constitution.md	R-23 mitigation — written, not merely planned (§6)

SpecKit conventions were followed for the specs /<###-feature>/ {spec,plan,tasks}.md layout and the template section structure. **create-new-feature.sh was deliberately not run**: it creates a git branch, and branch creation in a shared tree with eleven live agents is out of this task's remit. The directory was created at the path the script would have chosen.

2. Corrections to inherited framings

Three claims arrived with the Phase-2 brief. Each was checked against the artifact (§11.4.6), and two needed correcting. Recorded here because a plan built on a mis-framing inherits the mis-framing.

2.1 “The upstream repo is nearly empty of implementation” — PARTIALLY WRONG

Measured (02 §1, §4, §11): 237 Go files · 1 418 top-level declarations · 755 **exported symbols** · 26 packages · 13 extension-point interfaces · 12 migrations · 13 MCP tools · 27 CLI subcommands · **0 parse errors**. That is a production-shaped service, not an empty repository.

What is true is the narrower and sharper statement, and it is the one the plan uses:

- **0 .go files outside docs/** — every one is under docs/research/mvp/Agent_AI_Skill_Tree_Development/project/;
- **no root go.mod**, so go get against the repository URL cannot resolve it;
- the declared module path github.com/helixdevelopment/skill-system **disagrees with the repository URL** github.com/HelixDevelopment/skills in *both name and case*;
- **no release tags**, so any pin must be a raw SHA.

Why the distinction changes the plan. “Nearly empty” implies *write it*. The measurement says *relocate and reconcile it* — a smaller, safer operation that preserves 755 exported symbols instead of reinventing them. The brief's operative conclusions survive intact (**extend, not reuse; raw-SHA pin**); only the size of the work changes, and it shrinks.

06's R-01 heading (“ships nothing to incorporate”) is consistent with this once read precisely: nothing *consumable at a resolvable module path*. Phase 2 states it that way to remove the ambiguity.

2.2 “2 of 7 Bridge layers exist” — SUPERSEDED mid-phase

A dedicated layer-health probe landed during this phase with a materially different and better-resolved result:

WIRED: L1, L2, L6 · PARTIAL: L7 · DESIGNED-ONLY: L3, L4 · NONEXISTENT: L5.

The load-bearing change is that **the remedy is mixed, not uniform** — three absences need three different responses, and a uniform installer would necessarily contain a step that can never succeed:

Class	Layers	Correct response
ABSENT-BUT-OBTAINABLE	L3, L4	An installer genuinely fixes these — but must use an explicit --from release URL , because the bare specify extension add fails on a default install (the first-party catalogue holds 4 entries, all spec-kit-core; the bridges live in the community catalogue)
NONEXISTENT	L5	Delete the tier. No installer can help — there is nothing to install
ABSENT-BUT-BUILDABLE	L7	A build flag (-DGGML_RPC=ON; the submodule is already in-tree and already builds llama-server), not a procurement item

L6 was a false negative, and the mechanism matters more than the correction: `systemctl list-unit-files | grep -i helix` at **system** scope returns nothing because all seven Helix units are **user**-scope. A scope-limited probe produced a *confident* wrong answer — the §11.4.201 false-null class. FR-046 therefore requires the probe to enumerate both scopes, so the trap is **encoded rather than remembered**. This is logged as new risk **R-30**.

2.3 The work item's own environment facts are stale

HXC-159's PRE-VERIFIED ENVIRONMENT FACTS block states SpecKit “*is NOT yet initialized in this repo (no .specify/ and no specs/ directory present)*” and that “*Initialization is therefore an explicit early task.*” .specify/ **exists** with full scaffolding and ten registered speckit-* skills; the item's own later addendum says so. The tracker contradicts itself and **the addendum is correct**.

Consequence: **no L2 initialisation task is planned** (FR-045). Logged as **R-31**, with the general mitigation *re-derive environment facts from the artifact at phase start* (task T-P0.01) — because a plan that re-plans a completed step is working against a description instead of against the artifact.

3. The six open decisions, resolved

Ø3 §8 left six decisions to Phase 2. All six are taken in spec.md §2; summarised here with the evidence that forced each.

#	Decision	Forced by
D-1 Canonical schema = Agent Skills open standard ; the two proprietary dialects become profiled extensions with round-trip converters	It is now an open cross-vendor standard with ~45 implementations. Picking either dialect strands the other; picking the standard strands neither and yields §11.4.228 portability for free	
D-2 All three upstream layers in scope, ordered L1 → L3 → L2, with L2 never linked into a consumer binary	The internal/ visibility wall + R-25 (replace ignored outside the main module)	
D-3 Contract crosses the module boundary over the wire (MCP) plus a pure-data pkg/ loader	Ø3 §2/§6: pkg/ and the wire are the <i>only two proven</i> cross-module seams. Ø3b §5 calls the MCP-config path “the cleanest cross-consumer integration path in the entire analysis”	

#	Decision	Forced by
D-4 HelixAgent does not depend on HelixSkills — it becomes a client	Upstream's own helix-deps.yaml declares HelixAgent a dependency <i>of</i> HelixSkills. Inverting it risks a cycle CONST-051(C) forbids. Enforced mechanically by T-P6.05's module-graph gate	
D-5 Fix the two HelixCode structural defects first	No exported Skill constructor blocks the loader seam; the discarded CLI dispatcher kills auto-trigger on the exact surface the runtime signature is taken from	
D-6 The 1 174-file corpus does not migrate in this feature	Relocating it removes a tree from HelixAgent — a §11.4.122 event needing an operator <i>yes</i> before, not a report after. It stays vendored-tier and unactivated (which R-19 requires anyway)	

4. Sequencing rationale

Ten Track-A phases plus three Track-B phases, running concurrently and joining at P7.

The ordering is forced by measurement in four places, and each is easy to get wrong:

1. **P2 (shadowing benchmark) precedes every activation policy.** The published effect is established; its magnitude *on this corpus* is not. Setting a ceiling before the curve exists would be exactly the §11.4.6 guess this plan exists to avoid — so the benchmark is a **deliverable**, not an input. This is the

plan’s most schedule-surprising decision and the one the evidence forces most directly.

- 2. **P1 (hygiene) precedes P4 (loader).** The loader is fail-closed by design, so it will error immediately against today’s corpus — multitrack has a `register.sh` and no `SKILL.md`. Normalising first means P4’s first red is a real defect, not a known one.
- 3. **P3 (upstream extension) precedes P4.** R-01’s mitigation is explicit: “*an incorporation phase MUST NOT be scheduled before Phase C exists.*”
- 4. **P5 and P6 run in parallel, never sequential-copy.** Parity proven by two independent integrations is evidence; parity proven by copying one integration twice is a tautology.

Track B starts immediately because **no phase may declare a layer dependency before the real layer state is measured** — and because B3 (deleting L5) is a design correction that blocks nothing and unblocks the honest statement of what the Bridge is.

5. Risk coverage

All 29 register rows are dispositioned; none is left as a note. 26 open rows map to task groups with a named gate and a paired §1.1 mutation; 3 were already resolved in the register and are carried as design inputs (R-15 → pure-data skills; R-16 → R-23; R-17 → R-25). The full table is `plan.md` §7.

Three risks were found during planning and are new to the register:

ID	Risk	Sev	Mitigation	Task
R-29	Two independent installers provisioning into one <code>.claude/skills/tree</code> . Claude Code neither errors nor warns on a skill-name collision — it resolves silently — and the	HIGH	Install-time collision detection shipped by this feature: pre-write enumeration, abort-never-overwrite, reserved prefixes, manifest-driven regeneration. §11.4.8: NO external	T-P1.04

ID	Risk	Sev	Mitigation	Task
	upstream namespacing request was closed as not planned . The collision happens in the <i>filesystem</i> , before Claude Code reads the directory, so no upstream mechanism observes it		solution found – original work	
R-30	Scope-limited probes yield <i>confident</i> false negatives (the L6 case, §2.2)	MED	Every presence probe enumerates all applicable scopes and records which it checked — encoded in the probe, not remembered	T-B1.01
R-31	The work item's own environment facts are stale (§2.3)	MED	Re-derive environment facts from the artifact at phase start; record contradictions rather than reconciling them silently	T-P0.01

The risk register itself (06) is **not** edited by this document — it is a Phase-1b artefact at Revision 2, and amending a sibling agent's committed deliverable in a shared tree is not this task's remit. R-29/R-30/R-31 live in `plan.md` §7 and are proposed for a future Revision 3.

6. What was written vs planned

Planned only (no production code touched): everything in specs/.

Written: .specify/memory/constitution.md — replaced SpecKit's stock 2 346-byte placeholder with a CONST-059 inheritance pointer to constitution/Constitution.md.

Rationale for acting rather than scheduling: the file is a **governance pointer, not production code**, and the placeholder was **live** — read by every /speckit-plan Constitution Check run, including this feature's own. Leaving it would have allowed a generic template to shadow the real constitution during the very workflow this specification mandates. The unmodified SpecKit template remains untouched at .specify/templates/constitution-template.md, and the gate that makes the fix durable (CM-CANONICAL-ROOT-CLARITY extension) is scheduled as T-P1.06.4 rather than claimed.

No other file outside specs/ and this research tree was modified. No submodule pointer was touched. No branch was created. Nothing was pushed. docs/Issues.md, docs/Fixed.md and the summaries were **not** regenerated, per the coordinator's standing deferral.

7. Honest boundary

Verified: every factual claim traces to a cited section of the Phase-1 research tree or to a probe captured in this session. Two runtime observables were verified directly so the signature registry names real commands rather than plausible ones: helix_code/cmd/cli/main.go:3306 (the skills subcommand dispatcher) and submodules/helix_agent/internal/router/router.go:1363-1368 (GET /v1/skills, /categories, /:category, POST /match).

Not verified — nothing was built, run or wired. This is a planning phase. Under §11.4.108 every claim here is SOURCE-layer at best; no runtime signature in the registry has been observed, no gate exists, and no mitigation has been proven. None of them is “done” until verified on a clean target.

Not read: the six Bridge documents Phase 1 also left unread (NANO_TASK_ENGINE, EXTENSION_DEVELOPMENT, TDD_INTEGRATION, CONSTITUTION_INTEGRATION, SECURITY, APPENDIX). Obligations stated only there are not reflected in FR-044...FR-050 — task T-B2.04 closes this and **must complete before the installer is considered final** (U-7).

Nine UNCONFIRMED: items are carried in spec .md §12, each with what would settle it. Two were resolved during this phase (U-1 → L6 is WIRED; U-2 → no longer load-bearing because L5 is deleted rather than substituted). Seven remain open: the local shadowing magnitude (**U-3 — blocking; no threshold may be chosen before it**), the GitLab group enumeration (U-4), HelixAgent's unsurveyed Toolkit/ and MCP/ trees (U-5), CONST-040 capability sourcing (U-6), the unread Bridge documents (U-7), the four stale upstream branch diffs (U-8), and the upstream module's true maturity (U-9).

Where the plan is weakest: P2's cost is genuinely unestimable before its task set exists; U-5 could reshape P6's attachment point; and assumption A-2 (upstream is writable by this team) is an assumption, not a verified fact — if it fails, P3 becomes a mirror and the plan's shape changes at its root.

Sources

Phase-1 tree, this directory: 01_speckit_init.md ·
02_upstream_feature_inventory.md ·
03_consumer_integration_surfaces.md · 03b_extension_integration.md ·
04_deep_research.md · 05_catalogue_survey.md · 06_risk_register.md.

Local verification, 2026-07-29: sqlite3 docs/workable_items.db (HXC-159 description) · .specify/{templates,memory,scripts} inspection ·
helix_code/cmd/cli/main.go · submodules/helix_agent/internal/
router/router.go.

Bridge layer-health probe, 2026-07-29 (sibling stream): per-layer WIRED/PARTIAL/DESIGNED-ONLY/NONEXISTENT verdicts, supply-chain measurements for SuperSpec / SuperB / speckit-community / erophames/superpowers-mcp, and the spec-kit catalogue-scope finding.